

C612

Mathematics: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student teacher at a high school with a new STEM wing models a cart's velocity as $v(t) = t - 2$ meters per second for $0 \leq t \leq 4$. The cart was built by the robotics club. What is the cart's displacement over the 4 seconds?

- A. 8 m
- B. 4 m
- C. 2 m
- D. 0 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A curriculum coordinator who taught calculus for 15 years asks for the average value of $f(x) = x^2$ on the interval $[0, 3]$. The request came in an email sent at 6 a.m. What is the average value?

- A. 6
- B. 9
- C. 3
- D. 4.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An analyst at a logistics company with 18 warehouses uses the matrix A with rows (2, 4) and (3, k). The system worked well last quarter. For what value of k is A singular?

- A. $k = 6$
- B. $k = 1.5$
- C. $k = 0$
- D. $k = -6$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A 3×3 matrix A has determinant 5. Which statement is true?

- A. $\det(2A) = 10$
- B. A must have a row of zeros.
- C. $Ax = 0$ has infinitely many solutions.
- D. A is invertible, and $Ax = b$ has exactly one solution for every b .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher who coaches the math league gives students A , a 2×3 matrix, and B , a 3×4 matrix, in a unit that runs six weeks. Which statement is true?

- A. AB is 2×4 , and BA is undefined.
- B. AB and BA are both defined and equal.
- C. AB is 3×3 .
- D. AB is 4×2 .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A screening test detects a condition in 90% of people who have it and gives a false positive for 5% of people who do not. The condition affects 1% of the population. The test is sold in a bright blue box. If a person tests positive, about what is the probability they have the condition?

- A. 1%
- B. 90%
- C. About 15%
- D. 95%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

$P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$. What is $P(A | B)$?

- A. 0.5
- B. 0.4
- C. 0.7
- D. 0.2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A department chair at a school that won a state award must choose a 3-person committee from 10 teachers; all members have equal roles. The chair has served for four years. How many committees are possible?

- A. 720
- B. 120
- C. 1,000
- D. 30

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

From 10 club members, how many ways can a president, vice president, and secretary be chosen?

- A. 120
- B. 1,000
- C. 30
- D. 720

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A geometry teacher who writes the department newsletter posts: “If a quadrilateral is a square, then it is a rectangle.” The newsletter goes to 40 families. Which statement is logically equivalent?

- A. If a quadrilateral is not a square, then it is not a rectangle.
- B. If a quadrilateral is not a rectangle, then it is not a square.
- C. A quadrilateral is a square if and only if it is a rectangle.
- D. If a quadrilateral is a rectangle, then it is a square.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

To prove “if n^2 is even, then n is even” by contrapositive, what do you assume and show?

- A. Assume n is odd and show n^2 is odd.
- B. Assume n is even and show n^2 is even.
- C. Check $n = 2, 4$, and 6 .
- D. Assume n^2 is even and n is odd, then find a contradiction.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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12.

On a statewide exam with 40,000 test takers, scores have a mean of 70 and a standard deviation of 8. A student at a rural school scores 86. What is the student's z-score?

- A.** 0.5
- B.** 10.75
- C.** 2
- D.** 16

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A student teacher ~~at a high school with a new STEM wing~~ models a cart's velocity as $v(t) = t - 2$ meters per second for $0 \leq t \leq 4$. ~~The cart was built by the robotics club.~~ What is the cart's displacement over the 4 seconds?

Target word: displacement (net change) **Decides:** The integral of v from 0 to 4 is signed area.

A	Wrong stem number	Evaluates only $t^2/2$ at $t = 4$.
B	Adjacent category	That is total distance: two triangles of area 2.
C	Confusion set	That is the velocity at $t = 4$, not displacement.
D	Correct	$\int (t - 2) dt$ from 0 to 4 = $[t^2/2 - 2t] = 8 - 8 = 0$.

2. Answer C Concept 2

A curriculum coordinator ~~who taught calculus for 15 years~~ asks for the average value of $f(x) = x^2$ on the interval $[0, 3]$. ~~The request came in an email sent at 6 a.m.~~ What is the average value?

Target word: average value **Decides:** $(1/3) \int_0^3 x^2 dx$.

A	Confusion set	That is $f'(3)$, a derivative, not an average.
B	Dropped element	Forgets to divide by the interval length 3.
C	Correct	$\int_0^3 x^2 dx = 9$; $9 \div 3 = 3$.
D	Adjacent category	Averages the endpoint values 0 and 9 instead of the function.

3. Answer A Concept 3

An analyst at a logistics company ~~with 18 warehouses~~ uses the matrix A with rows $(2, 4)$ and $(3, k)$. ~~The system worked well last quarter.~~ For what value of k is A singular?

Target word: k making $\det = 0$ **Decides:** $\det = 2k - 12$.

A	Correct	$2k - 4 \cdot 3 = 0$ gives $k = 6$.
B	Confusion set	Pairs the wrong entries: $2 \cdot 3 - 4k = 0$.
C	Absolute language	A zero entry does not make a matrix singular; the determinant must be zero.
D	Inverted operation	Sign error: sets $2k + 12 = 0$.

4. Answer D Concept 3 · transfer

A 3×3 matrix A has determinant 5. Which statement is true?

Target word: what a nonzero determinant implies **Decides:** $\det \neq 0$ means invertible.

A	Wrong stem number	For 3×3 , $\det(2A) = 2^3 \times 5 = 40$.
B	Confusion set	A zero row would force $\det = 0$.
C	Inverted operation	That is true only when $\det = 0$.
D	Correct	A nonzero determinant guarantees invertibility.

5. Answer A Concept 4

A teacher ~~who coaches the math league~~ gives students A, a 2×3 matrix, and B, a 3×4 matrix, ~~in a unit that runs six weeks~~. Which statement is true?

Target word: product dimensions **Decides:** Inner dimensions must match; outer give the size.

A	Correct	$(2 \times 3)(3 \times 4) = 2 \times 4$; $(3 \times 4)(2 \times 3)$ has $4 \neq 2$.
B	Absolute language	Matrix products need not both exist and usually do not commute.
C	Confusion set	Uses the inner dimensions as the size.
D	Inverted operation	Reverses the outer dimensions.

6. Answer C Concept 5

A screening test detects a condition in 90% of people who have it and gives a false positive for 5% of people who do not. The condition affects 1% of the population. ~~The test is sold in a bright blue box.~~ If a person tests positive, about what is the probability they have the condition?

Target word: $P(\text{condition} \mid \text{positive})$ **Decides:** Rare prevalence makes false positives dominate.

A	Dropped element	The prior prevalence, ignoring the positive result.
B	Inverted operation	That is $P(\text{positive} \mid \text{condition})$, the reverse conditional.
C	Correct	$0.009 / (0.009 + 0.0495) = 0.009 / 0.0585 \approx 0.154$.
D	Confusion set	The true-negative rate, which answers a different question.

7. Answer B Concept 5 · transfer

$P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$. What is $P(A \mid B)$?

Target word: conditional probability **Decides:** $P(A \text{ and } B) / P(B)$.

A	Inverted operation	That is $P(B \mid A) = 0.2 / 0.4$.
B	Correct	$0.2 / 0.5 = 0.4$.
C	Confusion set	That is $P(A \text{ or } B) = 0.4 + 0.5 - 0.2$.
D	Dropped element	The joint probability, not divided by $P(B)$.

8. Answer B Concept 6

A department chair ~~at a school that won a state award~~ must choose a 3-person committee from 10 teachers; all members have equal roles. ~~The chair has served for four years.~~ How many committees are possible?

Target word: number of unordered groups **Decides:** Equal roles: combinations.

A	Adjacent category	$P(10, 3)$ counts orderings; roles are equal.
B	Correct	$C(10, 3) = 10 \cdot 9 \cdot 8 / 6 = 120$.
C	Dropped element	10^3 drops the no-repeat condition.
D	Wrong stem number	Multiplies the stem numbers 10×3 .

9. Answer D Concept 6 · transfer

From 10 club members, how many ways can a president, vice president, and secretary be chosen?

Target word: ordered selection **Decides:** Distinct roles: permutations.

A	Adjacent category	Combinations ignore the distinct roles.
B	Dropped element	Allows one person to hold several offices.
C	Wrong stem number	Multiplies 10×3 .
D	Correct	$10 \cdot 9 \cdot 8 = 720$.

10. Answer B Concept 7

A geometry teacher ~~who writes the department newsletter~~ posts: "If a quadrilateral is a square, then it is a rectangle." ~~The newsletter goes to 40 families.~~ Which statement is logically equivalent?

Target word: equivalent statement **Decides:** The contrapositive.

A	Confusion set	The inverse; equivalent to the converse, not the original.
B	Correct	The contrapositive always has the same truth value.
C	Adjacent category	A biconditional claims both directions.
D	Inverted operation	The converse; false for non-square rectangles.

11. Answer A Concept 7 · transfer

To prove "if n^2 is even, then n is even" by contrapositive, what do you assume and show?

Target word: setup of contrapositive proof **Decides:** Assume not-Q, show not-P.

A	Correct	Not-Q (n odd) leads to not-P (n^2 odd).
B	Inverted operation	That proves the converse.
C	Dropped element	Examples do not prove a general statement.
D	Adjacent category	That is proof by contradiction, a different method.

12. Answer C Concept 8

On a statewide exam ~~with 40,000 test takers~~, scores have a mean of 70 and a standard deviation of 8. A student ~~at a rural school~~ scores 86. What is the student's z-score?

Target word: z-score **Decides:** $(86 - 70) / 8$.

A	Inverted operation	Divides the SD by the difference: $8 / 16$.
B	Confusion set	Divides the raw score by the SD: $86 / 8$.
C	Correct	$16 / 8 = 2$.
D	Dropped element	Forgets to divide by the standard deviation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Definite integral as net change	The definite integral of a rate gives net change (signed area); total distance needs the integral of the absolute value.	Treating net displacement and total distance as the same.	A velocity that changes sign on the interval.
2	Average value of a function	The average value of f on $[a, b]$ is $(1/(b - a))$ times the integral of f from a to b .	Averaging the endpoint values, or forgetting to divide by the interval length.	"Average value" of a continuous function over an interval.
3	Determinant and invertibility	A square matrix is invertible exactly when its determinant is nonzero; then $Ax = b$ has a unique solution. For $n \times n$, $\det(kA) = k^n \det(A)$.	Thinking a zero entry makes a matrix singular, or that $\det(kA) = k \det(A)$.	"For what value is the matrix singular?"
4	Matrix multiplication	An $m \times n$ matrix times an $n \times p$ matrix gives $m \times p$; the inner dimensions must match, and AB usually does not equal BA .	Assuming matrix products commute or using the inner dimensions for the size.	Two matrices with different dimensions.
5	Conditional probability	$P(A B) = P(A \text{ and } B) / P(B)$; with a rare condition, most positive tests can be false positives.	Swapping $P(A B)$ and $P(B A)$.	A test with sensitivity, false-positive rate, and prevalence.
6	Permutations vs. combinations	Use permutations (nPr) when order or distinct roles matter and combinations (nCr) when only the group matters.	Using permutations for an unordered committee.	Titled roles vs. an unlabeled group.
7	Conditional logic and proof	A conditional is equivalent to its contrapositive, not its converse or inverse; proof by contrapositive assumes not- Q and shows not- P .	Treating the converse as equivalent, or confusing contrapositive with contradiction.	"Which statement is logically equivalent?"
8	Standard scores	$z = (x - \mu)/\sigma$ tells how many standard deviations a value is from the mean; about 95% of a normal distribution lies within $z = \pm 2$.	Forgetting to divide by the standard deviation.	A score with a mean and standard deviation.

Course scope (public description): Mathematics: Content Knowledge is designed to help candidates refine and integrate the mathematics content knowledge and skills necessary to become successful secondary mathematics teachers. A high level of mathematical reasoning skills and the ability to solve problems are necessary to complete this course. Prerequisites for this course are College Geometry, Probability and Statistics I, Pre-Calculus, Calculus I, and Calculus II. Linear Algebra, and Calculus III are recommended.